

Information Integration and Analytic Data Processing

2025/2026

Project description

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Designing a Domain-Specific Search Engine Using a Data Warehouse

Project Goal

The goal of this project is to design and implement a **search engine for a specific domain**, where:

- Data from multiple sources is **integrated into a data warehouse**
- The warehouse acts as the **central structured repository**
- The **information retrieval system searches and ranks information extracted from the warehouse**

Students must integrate **data engineering + data warehousing + information retrieval techniques**.

Project Overview

The project consists of **two main phases**:

Phase 1

Data Selection and Data Warehouse Construction

Select a domain, collect datasets, and integrate them into a data warehouse.

Phase 2

Information Retrieval System

Build a search engine that retrieves and ranks information derived from the warehouse.

Phase 1: Domain Selection and Data Warehouse

Step 1: Select a Domain

Domain:

Students must choose a **specific field of study or application area** for which a search engine will be built.

Examples include:

- Scientific publications
- Movies and entertainment
- News articles
- E-commerce products
- Legal documents
- Music databases
- Tourism information
- Healthcare research
- Video game databases
- Academic courses

The chosen domain must contain **both structured attributes and textual content** suitable for search.

Students must justify:

- Why a search engine is useful in this domain
- What type of users would use the system
- What types of queries the system should support

Show in class:

- **Domain description**

Dataset Identification

Students must identify **multiple datasets** that will be integrated into a single data warehouse.

Requirements:

Minimum datasets:

- **At least 2 datasets**
- At least **one must contain textual information**

Possible sources include:

- Open data portals
- Kaggle datasets
- APIs
- Web scraped data
- Public repositories

For each dataset students must document:

- Source
- Data format
- Number of records
- Key attributes
- Text fields available

- Data quality issues

Students must also explain **how the datasets relate to each other**.

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- **Dataset inventory report**

Integration Plan

Students must define how the datasets will be combined.

This includes:

- Identifying common identifiers
- Defining relationships
- Resolving schema differences
- Handling missing data
- Handling inconsistent naming

Students must describe:

- Integration challenges
- Data cleaning strategies
- Data normalization strategies

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- **Data integration plan**
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Step 2: Dimensional Modeling

Students must design a **dimensional model** suitable for analytical queries and search.

Requirements:

- Define the **grain of the fact table**
- Create **fact tables**
- Create **dimension tables**

Example dimensions might include:

- Time
- Author
- Product
- Category
- Organization
- Location

Students must provide:

- Star schema diagram
- Explanation of measures
- Description of each dimension

Show in class:

- **Dimensional model**
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Step 3: ETL Process

Students must implement an **ETL pipeline** to populate the data warehouse.

The ETL pipeline should include:

Extraction

Import data from original sources.

Transformation

Perform:

- Cleaning
- Data type conversion
- Text normalization
- Deduplication
- Integration of datasets

Loading

Insert transformed data into the warehouse tables.

Students should document:

- ETL workflow
- Transformation rules
- Error handling strategy

Tools may include:

- Python
- SQL
- Pandas
- Airflow
- dbt

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- **ETL description and implementation (eg.: CSV files with the fact table and dimensions)**
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Bonus: Physical Design of the Warehouse

Students must design and implement the **physical storage structure** of the warehouse.

Tasks include:

- Choosing a database system
 - Creating indexes
 - Implementing keys and constraints
 - Designing partitions if necessary
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Phase 2: Information Retrieval System

In this phase students develop a **search engine that retrieves information derived from the warehouse**.

Step 4: Extract Searchable Documents

Students must transform warehouse data into **searchable documents**.

Examples:

Movie database:

- Each movie becomes a document

Scientific articles:

- Each paper becomes a document

Products:

- Each product becomes a document

Documents may combine fields such as:

- title
- description
- category
- attributes
- reviews

Students must describe:

- Document structure
- Field weighting strategy

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- **Document generation description**
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Step 5: Text Representation and Preprocessing

Students must implement standard preprocessing techniques:

Required:

- Tokenization
- Stopword removal
- Stemming or lemmatization

Students must build:

- Term vocabulary
- Document representation

Possible representations:

- Bag-of-Words
- TF-IDF vectors

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- **Text preprocessing pipeline**
-

Step 6: Implement Retrieval Models

Students must implement at least **two retrieval models**.

Possible models:

- Boolean retrieval
- Vector Space Model
- BM25

Students must compare the results produced by each model.

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- **Search ranking implementation**
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Step 7: Evaluation of the Retrieval System

Students must evaluate the search engine using standard IR metrics.

Tasks:

1. Define a set of **test queries**
2. Create **relevance judgments**

3. Compute metrics such as:

- Precision
- Recall
- F1
- Precision@k
- nDCG
- MAP

Students must analyze the performance of their system.

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- **Evaluation report**
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Step 8: Learning to Rank or Neural Retrieval

Students must implement **one advanced retrieval approach**.

Possible options:

- Learning-to-rank model
- Embedding-based retrieval
- Neural ranking model
- Semantic search using transformer embeddings

Students must compare this approach with the classical retrieval models.

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- **Advanced retrieval experiment**
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Bonus: Conversational or Generative Search

Students can extend the system with a **conversational interface** or **generative search capability**.

Possible implementations:

- Natural language query interface
- Query rewriting
- Retrieval-Augmented Generation (RAG)

The system must use **retrieved documents from the warehouse-derived index** to generate answers.

Final Deliverables

Date: May 31st, 2026 (23:59)

Moodle

Zip file with code and report

Zip file name: IPAI2526_GroupNumber.zip, example IPAI2526_G01.zip

The report:

Maximum of 8 (eight) pages

Organized by steps

Discussion of points done/not done in the project

All group members should be identified in the report

AI Usage Policy for Student Projects

Students may use AI tools and platforms (such as ChatGPT, Copilot, Gemini, etc.) to assist with their projects. However, all students are fully responsible for understanding and being able to explain every part of the code, text, or information included in their work.

AI tools may not be used to generate an entire project or major sections of it. If a project appears to be fully AI-generated or if a student cannot explain their own code or content, the project will receive a grade of zero.

Students must be able to demonstrate and explain how their code or work functions.